Weibull reliability analysis of Formula 1 cars

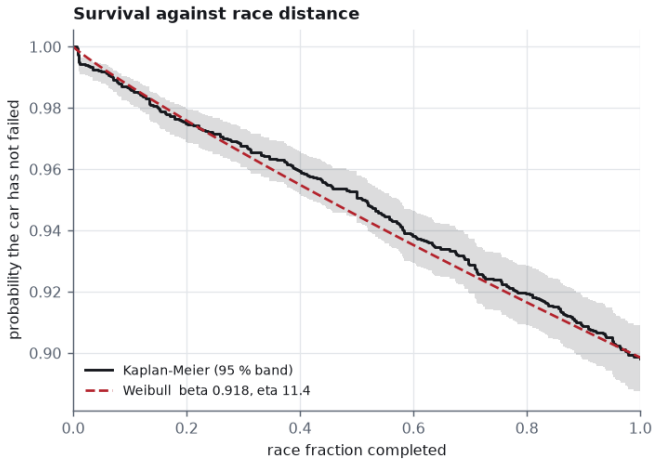
3,257 car-races across the 2014–2021 V6 hybrid era, 311 technical failures, 90.5% right-censored.

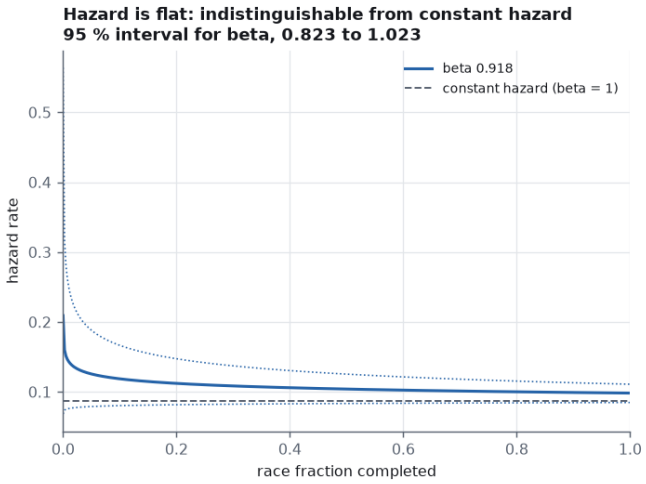
Beta is 0.918 — a constant hazard over a race distance

The 95% interval runs 0.823 to 1.023 and contains 1: an F1 car is no more likely to break in the last quarter of a race than in the first. A likelihood ratio against the exponential gives p = 0.117, so the simpler model cannot be rejected.

3,257	311	90.5%	0.918
car-races	technical failures	censored	shape, beta

Counting the finishers as failures instead — the commonest way to mishandle censored data — returns beta = 3.53, a confident wear-out signature that is not there.





Why 2014 to 2021

The later boundary is the feed, not a choice. From 2022 the data stops recording why a car retired, and a season that reports only "Retired" cannot separate an engine failure from a crash.

season	distinct statuses	named causes	rows with a named cause
2026	4	1	7
2024	5	2	5
2021	26	21	56
2018	28	23	83
2014	30	24	78

The earlier boundary is an engineering choice: 2014 opened the V6 turbo hybrid regulations, and a reliability model pooled across a regulation change is fitting two different machines at once.

What counts as a failure

A crash is not a reliability event. It is a competing risk: something else that can end the car's race first and so prevent the failure under study from ever being observed.

classification	car-races	share
technical	311	9.55%
accident	234	7.18%
driver	2	0.06%
operational	4	0.12%
unknown	38	1.17%
finished	2668	81.92%

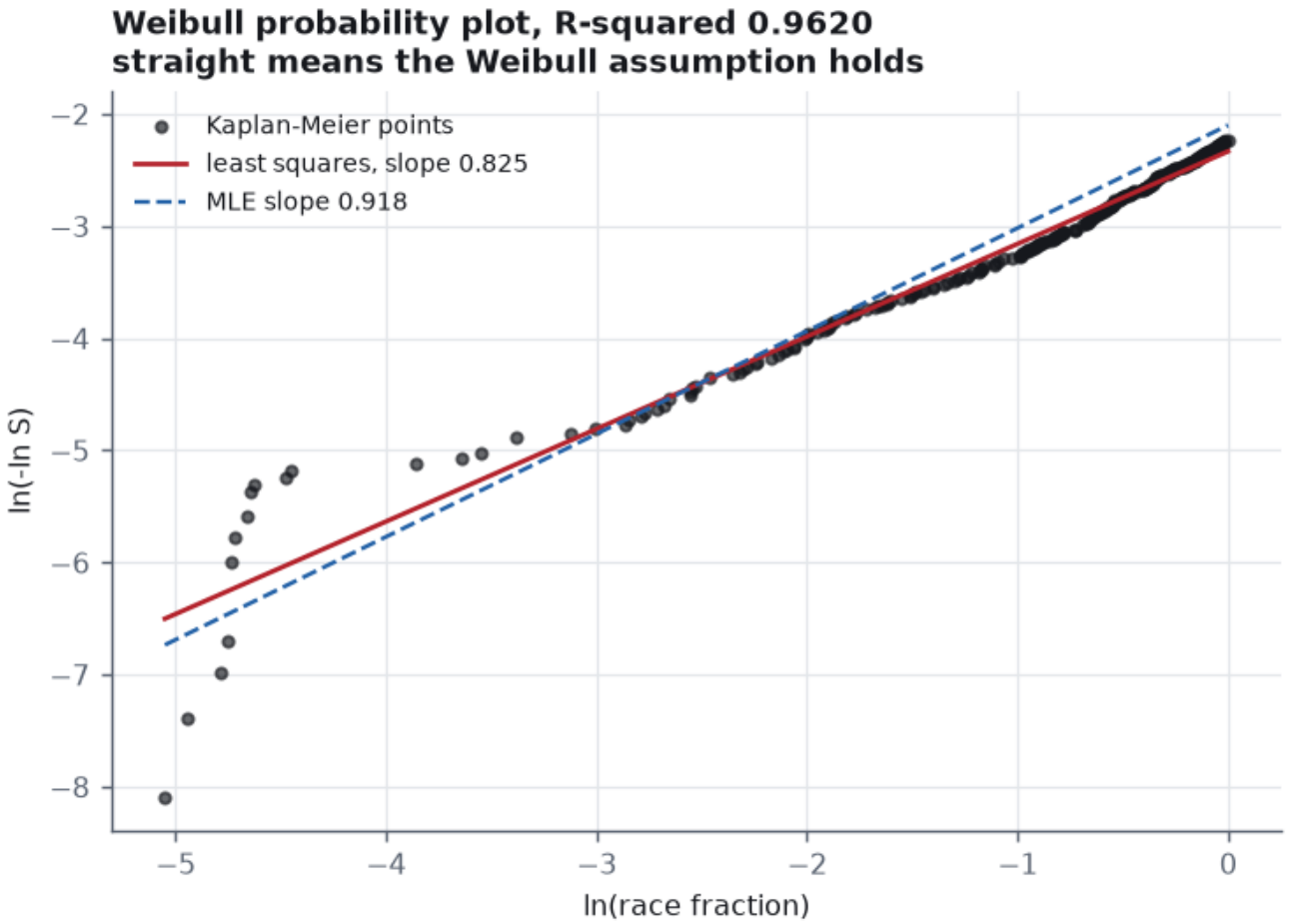
The fitted model

> eta is an extrapolation. The characteristic life comes out at 11.4 race distances, but no car in the data runs past one. The fit sees only the first 9% of that life, so eta and anything derived from it — the mean life above all — rest on the Weibull shape holding far beyond where it was observed. The B10 life of 0.983 sits at the edge of the data and is on much firmer ground.

quantity	value
shape, beta	0.9180 ± 0.0509
95% interval for beta	0.8235 – 1.0233
scale, eta (race distances)	11.412 ± 1.725
B10 life (race distances)	0.9833
survival to the flag, Kaplan-Meier	0.8983 (0.8870 – 0.9085)
restricted mean race fraction	0.9484
likelihood ratio against the exponential	chi-squared 2.45, p = 0.1172

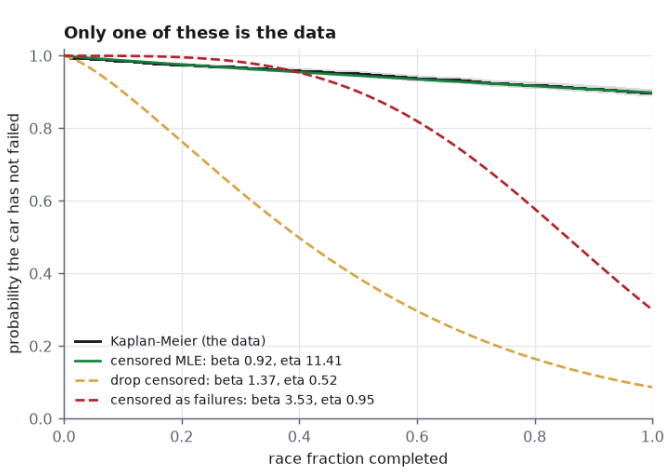
A check the fit does not grade itself on

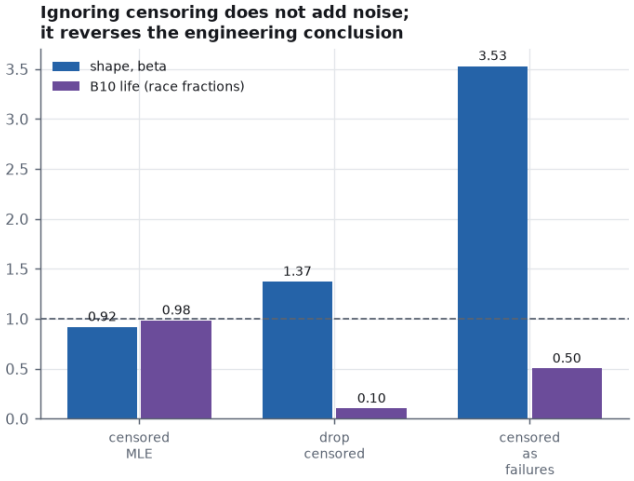
The transformed Kaplan-Meier points give a slope of 0.825 against the maximum likelihood 0.918, with R-squared 0.9620.



What censoring is worth

Neither wrong method is a rough approximation of the right one. Dropping the survivors throws away every long life and collapses the scale from 11.41 to 0.52; counting them as failures manufactures a population that nearly all dies at the same distance, which is what beta = 3.53 means.



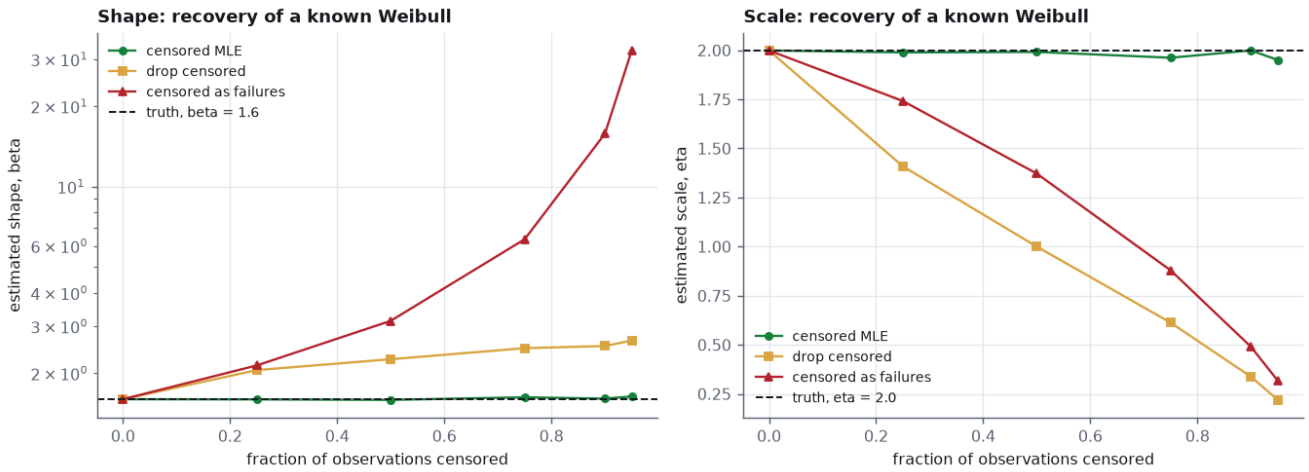


method	observations used	treated as failures	beta	eta	B10 life
censored MLE	3,257	311	0.918	11.412	0.983
drop censored	311	311	1.371	0.521	0.101
censored as failures	3,257	3,257	3.529	0.949	0.501

The same three methods on data whose answer is known

Simulated from a Weibull with beta = 1.6 and eta = 2.0.

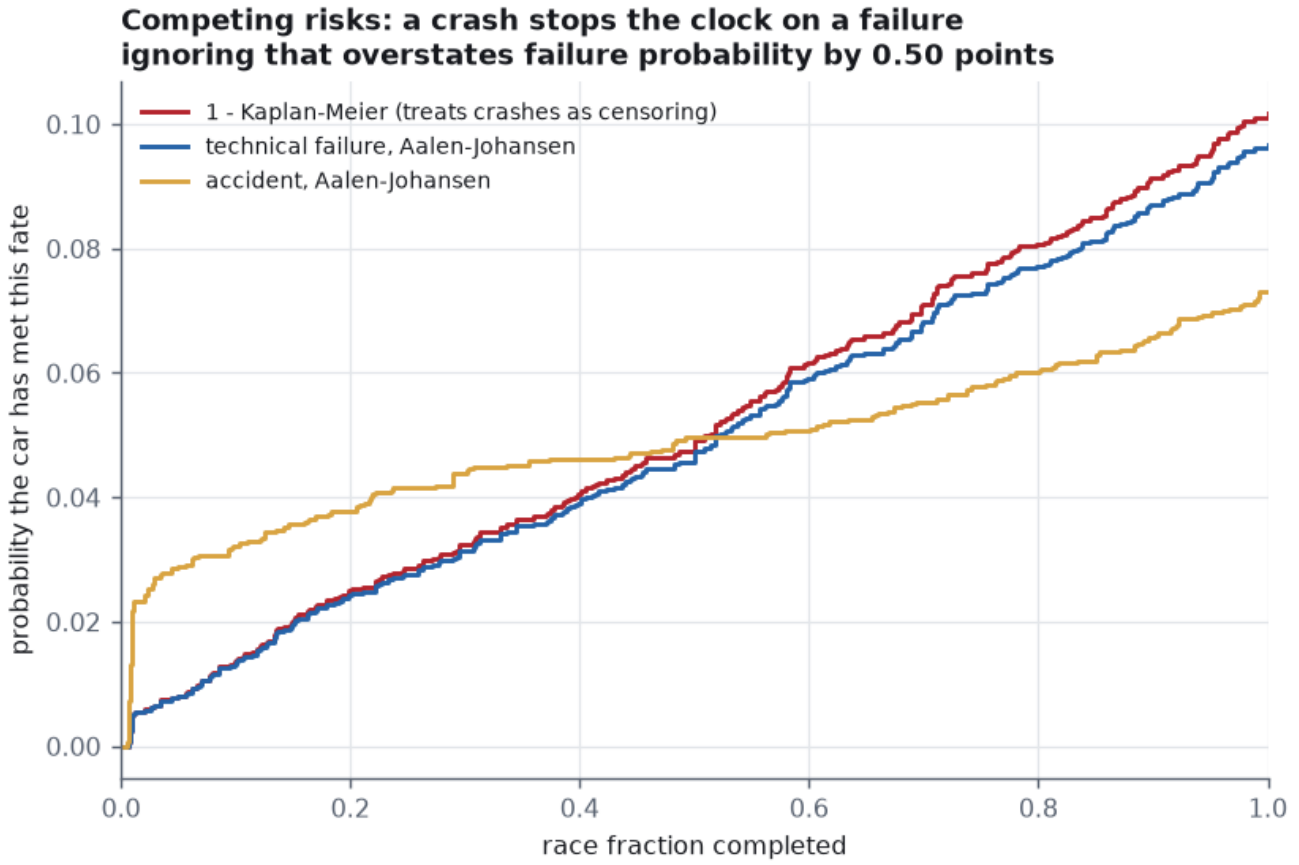
At zero censoring all three agree, as they must. At the 90% censoring of the real data the censored likelihood still recovers the truth and the other two do not.



censoring	MLE beta	MLE eta	dropped beta	dropped eta	as-failures beta	as-failures eta
0%	1.599	1.999	1.599	1.999	1.599	1.999
25%	1.596	1.989	2.055	1.409	2.140	1.742
50%	1.589	1.991	2.257	1.000	3.143	1.372
75%	1.625	1.962	2.483	0.613	6.352	0.879
90%	1.607	1.999	2.530	0.337	15.798	0.490
95%	1.637	1.950	2.651	0.219	32.395	0.317

Competing risks

Treating crashes as censoring assumes a car that crashes was just as likely to have failed later as one that did not — and then answers the question "what fraction would fail if crashes were abolished". The Aalen-Johansen estimator answers the question actually asked.

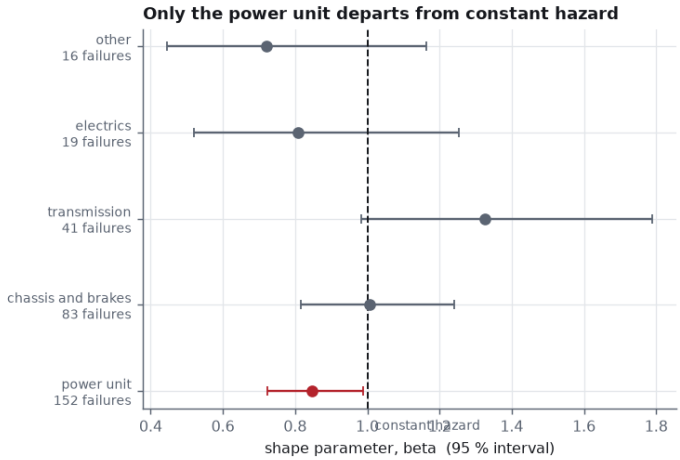


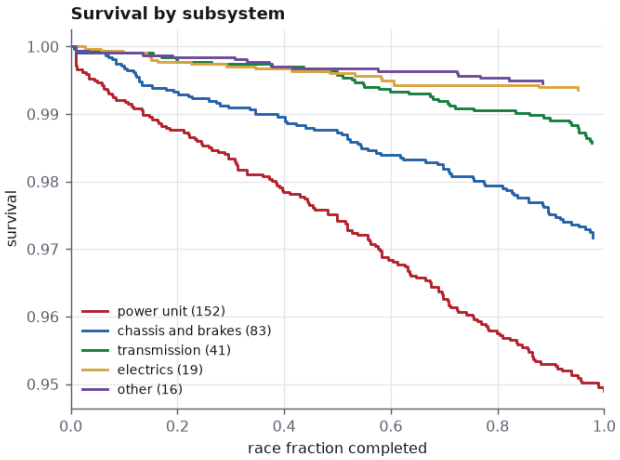
quantity	value
P(technical failure by the flag), Aalen-Johansen	0.0968
P(technical failure), naive 1 - Kaplan-Meier	0.1017
overstatement	0.50 points (1.051×)
P(accident by the flag)	0.0732

The correction is real but modest here, and saying so is part of the result: the bias grows with the competing event rate, and at a 7% crash rate there is not much of it to grow from.

By subsystem

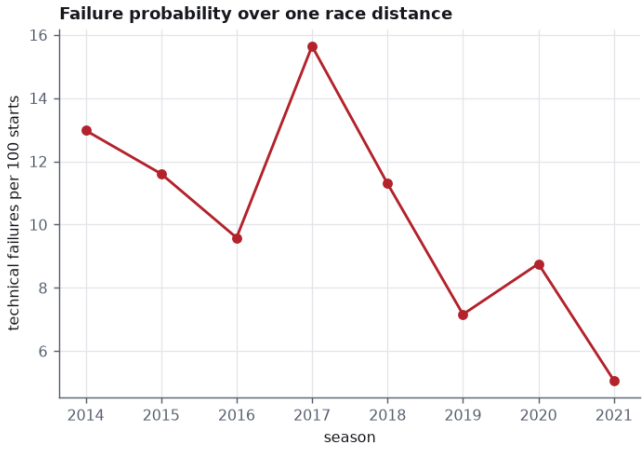
The pooled beta near 1 is a blend. The power unit is the one subsystem whose interval clears 1 entirely, and it does so on the infant-mortality side.

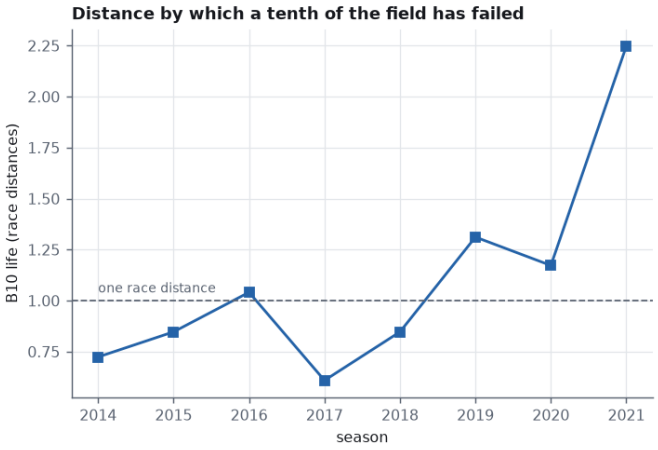




subsystem	failures	beta	95% interval	survival to the flag	verdict
power unit	152	0.846	0.724 – 0.989	0.9490	infant mortality
chassis and brakes	83	1.006	0.815 – 1.241	0.9718	constant hazard
transmission	41	1.326	0.983 – 1.789	0.9857	constant hazard
electrics	19	0.807	0.520 – 1.253	0.9936	constant hazard
other	16	0.720	0.446 – 1.162	0.9946	constant hazard

Reliability growth across the era

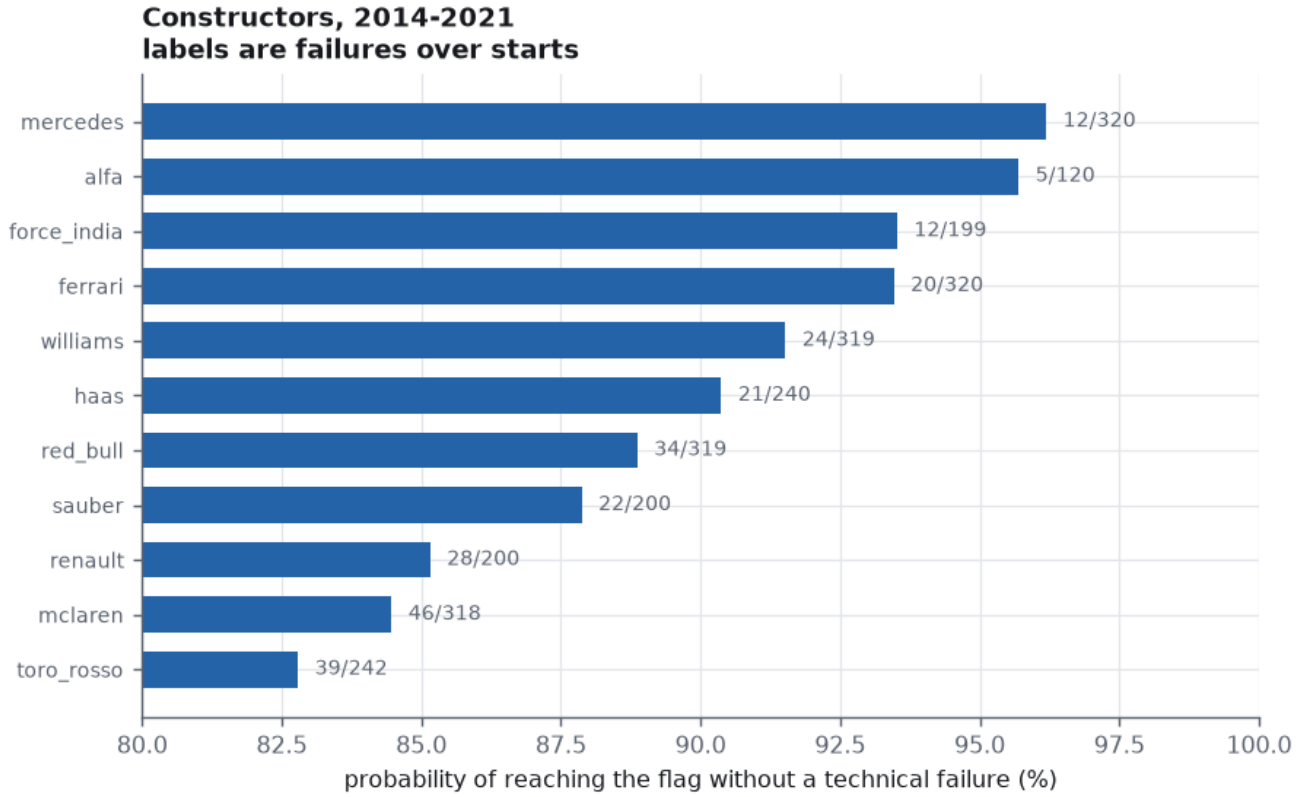




season	starts	failures	failure probability per race	beta	B10 life
2014	405	50	12.979%	0.885	0.724
2015	375	40	11.607%	0.826	0.847
2016	460	41	9.586%	1.026	1.041
2017	400	58	15.645%	0.966	0.609
2018	420	45	11.297%	0.816	0.847
2019	419	28	7.161%	1.382	1.311
2020	339	28	8.761%	0.839	1.172
2021	439	21	5.073%	0.879	2.248

By constructor

Log-rank, mercedes against toro_rosso: chi-squared 27.78 on 1 degree of freedom, p = 1.359e-07, with 12 failures observed against 30.5 expected.



constructor	starts	failures	survival to the flag	restricted mean
mercedes	320	12	0.9619	0.9796
alfa	120	5	0.9570	0.9811
force_india	199	12	0.9352	0.9726
ferrari	320	20	0.9346	0.9588
williams	319	24	0.9152	0.9724
haas	240	21	0.9037	0.9576
red_bull	319	34	0.8887	0.9359
sauber	200	22	0.8787	0.9448
renault	200	28	0.8516	0.9144
mclaren	318	46	0.8447	0.9305
toro_rosso	242	39	0.8279	0.9164

The thirty-eight unexplained retirements

Neither choice moves the shape parameter past the third decimal, so nothing in this analysis rests on it.

treatment	failures	beta	eta	survival to the flag
censored (default)	311	0.9180	11.412	0.8983
failures	349	0.9196	10.024	0.8866